Problem 1.1: Probability density and current

The time-dependent Schrödinger equation in 3D is

$$i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x},t) = -\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{x},t) + V(\mathbf{x})\psi(\mathbf{x},t).$$

Using the Schrödinger equation, we can calculate the probability density $\rho(\mathbf{x},t) = |\psi(\mathbf{x},t)|^2$, which gives the probability of finding the particle at an infinitesimally small region around $\mathbf{x}$ at time t. For a closed system, the probability density should satisfy the continuity equation:

$$\frac{\partial\rho(\mathbf{x},t)}{\partial t} + \nabla\cdot\mathbf{j}(\mathbf{x},t) = 0.$$

This is a statement of local conservation of probability, where $\mathbf{j}(\mathbf{x},t)$ is the probability current density (how much probability is flowing through a unit area per unit time at position $\mathbf{x}$ and time t).

a) Derive the standard expression for the probability current density:

$$\mathbf{j}(\mathbf{x},t) = \frac{\hbar}{2mi}\left(\psi^*(\mathbf{x},t)\nabla\psi(\mathbf{x},t) - \psi(\mathbf{x},t)\nabla\psi^*(\mathbf{x},t)\right).$$

b) Rewrite $\mathbf{j}(\mathbf{x},t)$ in terms of the momentum operator $\hat{\mathbf{p}} = -i\hbar\nabla$.

c) In the Schrödinger picture and using Dirac notation, find an expression for the current operator $\hat{j}(\mathbf{x})$ in a basis-independent way. Is this operator time-dependent? What about its expectation value? (hint: recall that $\psi(\mathbf{x}) = \langle\mathbf{x}|\psi\rangle$ and $\psi^*(\mathbf{x}) = \langle\psi|\mathbf{x}\rangle$).

d) Now consider a polar form for the wave function: $\psi(\mathbf{x},t) = \sqrt{\rho(\mathbf{x},t)}e^{iS(\mathbf{x},t)/\hbar}$, where $\rho(\mathbf{x},t)$ is the probability density and $S(\mathbf{x},t)$ is the phase of the wave function. Show that the probability current density can be written as

$$\mathbf{j}(\mathbf{x},t) = \frac{\rho(\mathbf{x},t)}{m}\nabla S(\mathbf{x},t).$$

What does this mean for a *real* wavefunction $\psi(\mathbf{x})$?

e) For a plane-wave solution of the form $\psi(\mathbf{x},t) = \frac{1}{\sqrt{V}}e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}$, where V is some normalization volume (formally, $V\to\infty$ for a bona-fide plane wave), $\mathbf{k}$ is the wave vector, and ω is the angular frequency, compute the probability density $\rho(\mathbf{x},t)$. Is it time-dependent? How about the probability current density $\mathbf{j}(\mathbf{x},t)$?

Solution to Problem 1.1 This exercise is a straightforward application of Schrödinger's equation and some vector calculus.

a) Compute the time derivative of the probability density:

$$\frac{\partial\rho}{\partial t} = \frac{\partial}{\partial t}|\psi|^2 = \frac{\partial}{\partial t}(\psi^*\psi) = \psi^*\frac{\partial\psi}{\partial t} + \psi\frac{\partial\psi^*}{\partial t}. \tag{1}$$

We will need Schrödinger's equation and its complex conjugate:

$$\frac{\partial\psi}{\partial t} = \frac{1}{i\hbar}\left(-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi\right), \tag{2}$$

$$\frac{\partial\psi^*}{\partial t} = -\frac{1}{i\hbar}\left(-\frac{\hbar^2}{2m}\nabla^2\psi^* + V\psi^*\right). \tag{3}$$

Substituting these into the equation above,

$$\frac{\partial \rho}{\partial t} = \psi^* \left(\frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi \right) \right) + \psi \left(-\frac{1}{i\hbar} \left(-\frac{\hbar^2}{2m} \nabla^2 \psi^* + V\psi^* \right) \right) \tag{4}$$

$$= -\frac{\hbar}{2mi} \left(\psi^* \nabla^2 \psi - \psi \nabla^2 \psi^* \right) = -\nabla \cdot \left(\frac{\hbar}{2mi} \left(\psi^* \nabla \psi - \psi \nabla \psi^* \right) \right), \tag{5}$$

proving our result that

$$\mathbf{j}(\mathbf{x}, t) = \frac{\hbar}{2mi} \left(\psi^*(\mathbf{x}, t) \nabla \psi(\mathbf{x}, t) - \psi(\mathbf{x}, t) \nabla \psi^*(\mathbf{x}, t) \right).$$

b) This is a trivial substitution using $\hat{\mathbf{p}} \to -i\hbar\nabla$.

Okay, to be precise, we should think about how the momentum operator acts in Dirac notation:

$$\hat{p}\psi(\mathbf{x}, t) = \langle \mathbf{x} | \, \hat{\mathbf{p}} \, | \psi(t) \rangle = -i\hbar\nabla \langle \mathbf{x} | \psi(t) \rangle = -i\hbar\nabla\psi(\mathbf{x}, t). \tag{6}$$

where I am making clear that the operator $\hat{\mathbf{p}}$ acts on the ket $|\psi(t)\rangle$, while the "operator" ∇ acts on functions of position $\mathbf{x}$ (namely, the wave function $\psi(\mathbf{x}, t)$ in this case). *If this confuses, check S&N Section 1.7.2.* Therefore, the conjugate of this expression should be something like

$$\hat{p}\psi^*(\mathbf{x}, t) = \langle \psi(t) | \, \hat{\mathbf{p}} \, | \mathbf{x} \rangle = (\langle \mathbf{x} | \, \hat{\mathbf{p}} \, | \psi(t) \rangle)^* = (-i\hbar\nabla\psi(\mathbf{x}, t))^* = i\hbar\nabla\psi^*(\mathbf{x}, t). \tag{7}$$

Note how the sign changed. Also, note that we used the fact that the momentum operator is Hermitian, so $\hat{\mathbf{p}}^\dagger = \hat{\mathbf{p}}$ (it has to be, momenta are real valued eigenvalues).

Note: The operation $(-i\hbar\nabla)^\dagger$ is not really well defined because the $\dagger$ acts on operators in *Hilbert* space. You can imagine doing such an operation when ∇ appears in an inner product, for example $\langle \phi | \, \hat{p} \, | \psi \rangle = \int d^3x \, \phi^*(\mathbf{x})(-i\hbar\nabla\psi(\mathbf{x}))$. In that case, the $\dagger$ operation can be thought of telling the derivative to act on left, rather than on the right:

$$\langle \phi | \, \hat{p} \, | \psi \rangle = \int d^3x \, \phi^* \left(-i\hbar\nabla\psi \right) = \int d^3x \, (i\hbar\nabla\phi^*)\psi = \langle \phi | \, \hat{p}^\dagger \, | \psi \rangle. \tag{8}$$

where the middle equality comes from integrating by parts and assuming the wave functions vanish at infinity. Abusing notation a little bit, we can say that the derivative is an anti-Hermitian operator: "$\nabla^\dagger = -\nabla$" as long as we think of it as something that lives inside inner products: $\int f^* \nabla g \, d^3x = -\int (\nabla f)^* g \, d^3x$.

If we are okay a little bit of abuse of notation, we can write

$$\mathbf{j}(\mathbf{x}, t) = \frac{1}{2m} \left(\psi^*(\mathbf{x}, t) \hat{\mathbf{p}} \psi(\mathbf{x}, t) + \psi(\mathbf{x}, t) \hat{\mathbf{p}} \psi^*(\mathbf{x}, t) \right), \tag{9}$$

as long as we interpret this as meaning:

$$\mathbf{j}(\mathbf{x}, t) = \frac{1}{2m} \left(\langle \psi(t) | \mathbf{x} \rangle \langle \mathbf{x} | \, \hat{\mathbf{p}} \, | \psi(t) \rangle + \langle \psi(t) | \, \hat{\mathbf{p}} \, | \mathbf{x} \rangle \langle \mathbf{x} | \psi(t) \rangle \right). \tag{10}$$

c) Recall that the state $|\psi\rangle$ is basis-independent, and that when we write $\psi(\mathbf{x}) = \langle \mathbf{x} | \psi \rangle$, we are expressing the state in the position basis. One could just as well express it in the momentum basis, $\phi(\mathbf{p}) = \langle \mathbf{p} | \psi \rangle$, or any other basis for that matter. The action of some operator $\hat{O}$ on the state $|\psi\rangle$ is also basis-independent.

Using the result stated above (the careful version), we have

$$j(\mathbf{x}, t) = \frac{1}{2m} \Big(\langle \psi(t) | \mathbf{x} \rangle \langle \mathbf{x} | \, \hat{\mathbf{p}} \, | \psi(t) \rangle + \langle \psi(t) | \, \hat{\mathbf{p}} \, | \mathbf{x} \rangle \langle \mathbf{x} | \psi(t) \rangle \Big).$$

Technically, we are not done yet, as the questions asks for the current operator $\hat{j}(\mathbf{x})$ (note the lack of a time dependence). This is obtained by *unsandwiching* (definitely a word) the state $|\psi\rangle$:

$$\hat{j}(\mathbf{x}) = \frac{1}{2m}\Big(|\mathbf{x}\rangle \langle\mathbf{x}| \,\hat{\mathbf{p}} + \hat{\mathbf{p}} \,|\mathbf{x}\rangle \langle\mathbf{x}| \Big).$$

To be more precise, this *unsandwiching* operation is simply undoing the step of calculating the expectation value of the current operator, namely $j(\mathbf{x}, t) = \langle\psi(t)| \,\hat{j}(\mathbf{x})\, |\psi(t)\rangle$.

An important point here: the state $|\psi(t)\rangle = U(t, 0) \,|\psi(0)\rangle$ is time-dependent in the **Schrödinger picture**. It is obtained from the initial state by the unitary operator of time evolution $U(t, 0)$. Operators are not time-dependent in this picture (unless they have explicit time dependence).

d) You can show this by direct substitution. Sparing us both from the details, I will write the first and last step:

$$\mathbf{j}(\mathbf{x}, t) = \frac{\hbar}{2mi}\left(\sqrt{\rho}e^{-iS/\hbar}\nabla(\sqrt{\rho}e^{iS/\hbar}) - \sqrt{\rho}e^{iS/\hbar}\nabla(\sqrt{\rho}e^{-iS/\hbar})\right) = \frac{\rho}{m}\nabla S. \tag{11}$$

Note that we used the fact that $\nabla\rho = 2\sqrt{\rho}(\nabla\sqrt{\rho})$ and that $\nabla e^{iS/\hbar} = (i/\hbar)e^{iS/\hbar}\nabla S$.

Note that for a *real* wavefunction, the phase S is zero, so $\mathbf{j}(\mathbf{x}, t) = 0$ (we mean $\psi(\mathbf{x})$ is real for all $\mathbf{x}$, since the time-evolution phase does not even enter our expression for j). This implies the state is *stationary*. You already knew that from the simple observation that a real wavefunction must be an energy eigenstate with no time dependence other than a global phase.

This result is quite interesting: it shows that the flow of probability depends on the rate of change of the phase of the wave function. With an abuse of notation, we may even interpret $\nabla S/m$ as a velocity field for the probability flow, $j = \rho\mathbf{v}$. In the context of the WKB approximation, one can show that this phase is closely related to the classical action (Hamilton-Jacobi action), hence the notation S.

e) For the plane wave, the probability density is just a constant: $\rho(\mathbf{x}, t) = |A|^2$, which is independent of time, $\partial\rho/\partial t = 0$. The LHS of the continuity equation is zero, so the RHS must also be zero: $\nabla \cdot \mathbf{j} = 0$. But what is $\mathbf{j}$? Using the result above, we have that $\nabla S = \hbar\mathbf{k}$, so the probability current density is

$$\mathbf{j}(\mathbf{x}, t) = \frac{|A|^2}{m}\hbar\mathbf{k}.$$

Despite the fact that the probability density is constant in space and time, the probability density is constant flowing in the direction of the wave vector $\mathbf{k}$. This is consistent with our intuition of a plane wave representing a free particle with momentum $\mathbf{p} = \hbar\mathbf{k}$.